

Cultivar-dependent transcriptional responses of lychee (*Litchi chinensis*) to *Peronophythora litchii*: from exploratory candidate discovery to a prospectively registered genome-wide analysis with cross-context external evaluation

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Abstract

Litchi downy blight, caused by the oomycete *Peronophythora litchii*, is among the most damaging diseases of lychee (*Litchi chinensis* Sonn.), and public RNA-seq cohorts spanning cultivars, tissues, and infection time points offer a natural resource for identifying cultivar-dependent infection responses. Such data are easily over-interpreted, however, when the same cohort is used both to select candidates and to appear to confirm them. Here we unify two stages of a transcriptomic reanalysis of the lychee-*P. litchii* interaction into a single account. In the first, exploratory stage, we reanalyzed leaf RNA-seq data from the cultivars Guiwei and Yurong1 (GSE201243/PRJNA830488), identified 17 and 117 infection-responsive genes in the two cultivars, and prioritized 18 candidates with defense-associated annotations and apparently enriched stress- and hormone-associated promoter elements. In the second, confirmatory stage, we prospectively registered a discovery and external-evaluation protocol that fixed dataset roles, statistical models, thresholds, and evidence vocabulary before any external outcome was examined. A genome-wide negative-binomial cultivar-by-infection interaction analysis of 19,445 expressed genes produced 262 statistical candidates, of which 206 were retained after uniform mapping and gene-model quality control, 19 remained significant under an independent genome-wide edgeR quasi-likelihood analysis, and 16 passed the complete registered robustness procedure. In the locked cross-context evaluation against an independent pericarp time course contrasting a susceptible and a resistant cultivar (PRJNA450886), two genes were supported and five were directionally contradictory; the internally robust and externally supported sets did not overlap, an outcome compatible with chance given the set sizes. A formal audit showed that none of the original 18 exploratory candidates met the newly applied genome-wide interaction criterion and that two failed uniform-mappability control. Deterministic evidence integration assigned no gene to the highest confidence tier, placed 12 internally robust genes, whose annotations include carbohydrate and cell-wall remodeling, lipid-transfer, and chaperone-regulatory functions, in a middle tier, and retired 65 entities, with all null and contradictory outcomes reported. The unified analysis shows that candidates selected for effect-size prominence in a single cohort largely fail a formally specified genome-wide interaction test, delimits exactly which computational claims about this pathosystem the current public data can support, and releases a candidate resource whose evidence boundaries are explicit. Experimental perturbation remains necessary for any causal conclusion.

Keywords: *Litchi chinensis*; *Peronophythora litchii*; downy blight; RNA-seq; cultivar-by-infection interaction; preregistered analysis; cross-context evaluation

1 Introduction

Lychee (*Litchi chinensis* Sonn.) is a subtropical fruit crop of substantial economic importance in southern China, Southeast Asia, and a growing set of production regions worldwide [1]. Among its diseases, litchi downy blight, caused by the oomycete *Peronophythora litchii* (historically also treated as *Phytophthora*

litchii [2]), is particularly damaging: the pathogen attacks young leaves, inflorescences, and fruit, produces water-soaked brown lesions, and drives severe postharvest decay [2,3]. Because chemical control is costly and resistance breeding depends on understanding how host genotypes differ in their response to infection, the molecular basis of cultivar-level variation in the response to *P. litchii* is a question of both agronomic and biological interest.

The resources needed to address this question computationally have accumulated rapidly. High-quality genome assemblies are available for lychee [4], and several groups have profiled the transcriptional response to *P. litchii* infection. Sun et al. compared a susceptible and a resistant cultivar across a pericarp infection time course and reported that the two genotypes deploy distinct early response programs [5]. Liu et al. described calcium-dependent protein kinase (LcCDPK) family members induced by *P. litchii* inoculation [6], and Li et al. showed that the pathogen RXLR effector PIAvh202 promotes virulence by destabilizing a host ethylene-biosynthesis enzyme [7]. A pathogen-side transcriptome catalogued the *P. litchii* pathogenicity arsenal [2]. In parallel, public repositories now hold RNA-seq series covering resistant and susceptible cultivars, leaf and fruit tissues, messenger and small RNA, and multiple time points. These series offer an opportunity, in principle, to distinguish genome-wide cultivar-dependent infection responses from responses common to all genotypes.

Realizing that opportunity is statistically demanding. The quantity of interest is not differential expression within a cultivar but the cultivar-by-infection interaction: the difference between infection responses across genotypes. Interaction effects are estimated with less precision than main effects, the deposited designs are small (typically three libraries per cultivar-treatment cell), and a genome-wide search requires an explicit multiplicity family. Analyses that instead select a handful of strongly responding genes and then characterize them in the same cohort conflate selection with confirmation: large observed fold changes in a small experiment are a mixture of true signal and noise, and features chosen for extremity will regress toward null in any independent assessment. These concerns apply directly to the present pathosystem, in which most published candidate genes derive from single-cohort analyses without external evaluation.

Our own earlier work is part of that record. In an exploratory reanalysis of the GSE201243 leaf series [8] (Guiwei and Yurong1, mock versus inoculated, 24 h post-inoculation), we identified 17 significant infection-responsive genes in Guiwei and 117 in Yurong1, prioritized 18 candidates with defense-plausible annotations (among them jacalin-like lectin, pathogenesis-related and thaumatin/osmotin-like proteins, a heat-shock protein, lipid-transfer proteins, and calcium-binding proteins), and reported that a set of stress- and hormone-associated *cis*-regulatory elements was enriched in their promoters relative to a randomized background [9]. That analysis was framed as hypothesis-generating, and peer review of it converged on three substantive criticisms: the cultivar-by-treatment comparison was an effect-size contrast over pre-selected genes rather than a genome-wide interaction test with false-discovery-rate control; no independent dataset was used for validation; and promoter-element counting, even against a randomized background, is weak evidence of regulatory function. We regard all three criticisms as correct, and they motivated the confirmatory stage reported here.

This paper therefore presents the complete arc of the project in one place. We prospectively registered a discovery and external-evaluation protocol that locked dataset roles, models, thresholds, empty-result behavior, and the vocabulary used to describe evidence before any external outcome was opened. Because every cohort is public, this staged unlock is a computational safeguard enforced by the workflow rather than blinding in the experimental sense, and we document the registration chronology in the Methods. Within the locked discovery cohort we fitted a genome-wide negative-binomial interaction model, applied uniform mapping and gene-model quality control, and subjected the resulting candidates to a battery of internal robustness checks spanning independent quantification, an alternative statistical framework, filter sensitivity, and leave-one-library-out refits. We then evaluated the frozen candidates, a frozen pathway set, and a frozen

Table 1: Locked dataset roles and eligibility.

Accession	Design	Locked role	Evidence class
PRJNA830488 (GSE201243)	Guiwei/Yurong1 leaf; mock/infected; 24 h; 3 libraries per cell	Discovery	Conditional inferential
PRJNA450886	Guiwei/Heiye pericarp; mock/infected; 6, 24, 48 h; 3 libraries per cell	Primary external evaluation	Cross-context
PRJNA922966 (GSE222651)	Feizixiao leaf/fruit; mock/infected; 24 h; 3 libraries per cell	Generic infection transfer	Cross-context
PRJNA922965 (GSE222650)	Feizixiao leaf/fruit small RNA; mock/infected; 24 h; 3 libraries per cell	Orthogonal modality	Orthogonal
PRJNA1090613 (GSE262200)	Guiwei/SFZ leaf; mock/infected; 3 libraries per cell	Exploratory only	Exploratory

signed expression signature in independent public cohorts whose roles (primary cross-context evaluation, generic transfer, orthogonal modality, or exploratory) were fixed in advance. Orthogonal evidence classes (sequence annotation, promoter motifs, small-RNA coherence, and accession-aware literature) were kept separate from expression support, and a deterministic tier system integrated all layers while explicitly permitting an empty top tier. Finally, we audited the 18 exploratory candidates from the first stage against the genome-wide analysis. The result is a candidate resource for lychee downy blight whose every claim carries its evidence class, together with a case study in how much, and how little, of a plausible exploratory analysis survives formal statistical scrutiny.

2 Results

2.1 A two-stage design with locked dataset roles

The study comprises an exploratory stage completed first and a confirmatory stage whose protocol was registered before its analyses began (Figure 1A; Table 1). Five public cohorts enter the confirmatory stage in fixed, non-interchangeable roles. PRJNA830488 (GSE201243), the leaf series for Guiwei and Yurong1 at 24 h, is the sole discovery cohort; it is also the dataset analyzed in the exploratory stage. PRJNA450886, a pericarp time course contrasting the susceptible cultivar Guiwei with the resistant cultivar Heiye at 6, 24, and 48 h [5], serves as the primary external evaluation. Because it differs from the discovery cohort in tissue, resistant comparator, and time structure, we describe positive outcomes in it as *cross-context support* rather than replication. PRJNA922966 (Feizixiao leaf and fruit, mock versus infected) tests generic infection-response transfer but cannot address the cultivar interaction; PRJNA922965 provides small-RNA libraries as an orthogonal modality; and PRJNA1090613 (Guiwei versus SFZ leaf) is exploratory only, because its time and resistance metadata could not be resolved. Under the registered firewall, no external outcome could alter the frozen discovery membership, signature weights, or thresholds, and the protocol explicitly allowed the top evidence tier to be empty.

2.2 Stage 1: exploratory candidate discovery in GSE201243

The exploratory stage analyzed the 12 discovery libraries with per-cultivar differential-expression contrasts (infected versus mock) and identified 17 significant genes in Guiwei (15 up-regulated, 2 down-regulated) and 117 in Yurong1 (39 up-regulated, 78 down-regulated) at Benjamini–Hochberg adjusted $P < 0.05$, sug-

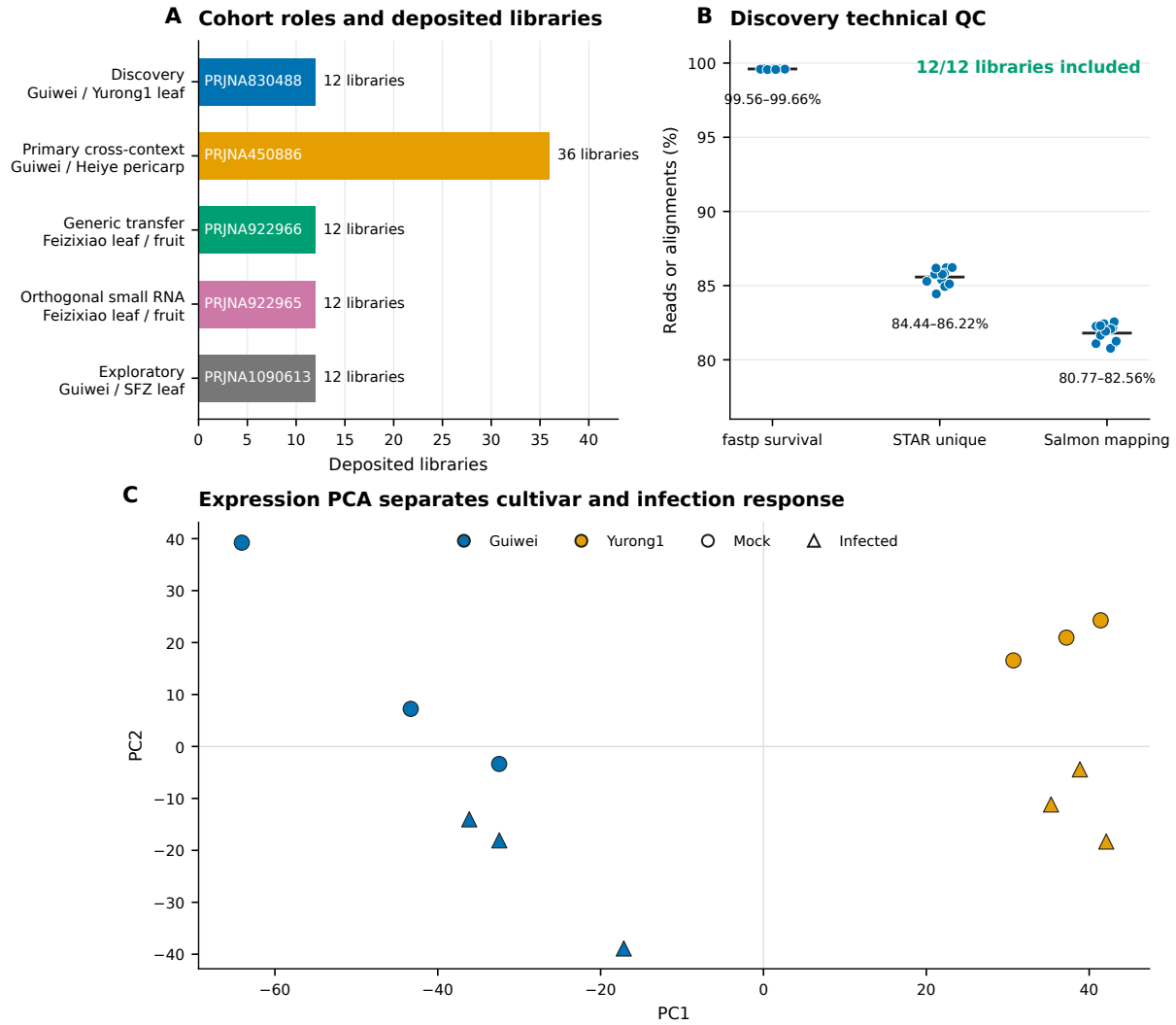


Figure 1: Study cohorts, discovery quality control, and expression structure. (A) The five public cohorts with locked roles and deposited library counts. (B) Per-library technical quality in the discovery cohort: fastp read survival, STAR unique-alignment rate, and Salmon mapping rate; all 12 libraries passed the fixed inclusion gates. (C) Principal-component analysis of the discovery libraries separates cultivar and infection status.

gesting a broader transcriptional response in Yurong1. From these lists, 18 genes were prioritized by combining statistical significance in at least one cultivar, large infection-associated fold changes, and annotations plausibly related to defense: among them a jacalin-like lectin (LITCHI028104), a pathogenesis-related protein (LITCHI011394), osmotin/thaumatin-like proteins (LITCHI009301, LITCHI002793), an HSP70-family chaperone (LITCHI012388), lipid-transfer proteins (LITCHI019183, LITCHI018043), an EF-hand calcium-binding protein (LITCHI019519), and a phytoalexin precursor (LITCHI007102). A selected-gene contrast of infection responses between cultivars nominated LITCHI019519, LITCHI001510, LITCHI028401, and LITCHI017676 as Guiwei-skewed and LITCHI028104 and LITCHI019183 as Yurong1-skewed candidates.

The exploratory stage also examined 2-kb upstream promoter regions of the 18 candidates. Against 100,000 randomized GC-matched promoter sets, four *cis*-element classes appeared enriched: anaerobic-response elements (ARE; 46 observed versus 24.6 expected, $q = 0.0003$), methyl-jasmonate-responsive motifs (54 versus 38.4, $q = 0.018$), TCA elements (9 versus 0.19, $q = 0.00003$), and TC-rich repeats (10 versus 0.15, $q = 0.00003$). ABRE and MYB-binding-site motifs, in contrast, were frequent but not enriched. All of these results were obtained by selecting and characterizing genes within a single cohort; they carry no genome-wide multiplicity control and no external evaluation, and we treat them below strictly as hypotheses submitted to the confirmatory stage. Full exploratory-stage figures, tables, and workflow scripts are available with the original report [9].

2.3 Discovery-cohort quality under the registered protocol

The confirmatory pipeline combined a joint host and pathogen reference, fastp trimming, STAR two-pass alignment, featureCounts gene counting, and independent Salmon quantification. Under this pipeline, all 12 discovery libraries passed the fixed technical gates of at least 10 million surviving read pairs and at least 40% uniquely aligned reads (Figure 1B). Read survival after trimming ranged from 99.56% to 99.66%, unique-alignment rates from 84.44% to 86.22%, and Salmon mapping rates from 80.77% to 82.56%. Principal-component analysis of the transformed counts separated libraries by cultivar on the first component and by infection status on the second, with no library isolated from its replicate group (Figure 1C). Because source-tree, pooling, and extraction independence are not reported for the deposited libraries, all inferential statements below are conditional on deposited-library independence, a limitation we return to in the Discussion.

2.4 Genome-wide cultivar-by-infection discovery

The primary discovery model fitted all expressed genes with the design $\sim$ cultivar + treatment + cultivar:treatment, with Guiwei and mock as reference levels, so that the interaction coefficient estimates how much the infected-minus-mock response in Yurong1 exceeds that in Guiwei. Of 19,445 expressed genes, 262 met the registered discovery threshold of genome-wide Benjamini–Hochberg $q < 0.05$ with an absolute interaction $\log_2$ fold change of at least $\log_2(1.5)$ (Figure 2A); 277 genes met the significance criterion alone and 5,370 met the effect criterion alone. All interaction estimates reported and plotted here are unshrunk maximum-likelihood values; apeglm shrinkage was used only to define the frozen signature weights. Uniform quality control then removed 31 candidates for exon-level mappability failure and 25 for gene-model ambiguity, freezing 206 genes as statistical candidates (Figure 2C). Because these quality filters were applied after significance testing, we verified that the ordering is not load-bearing: restricting the testing universe to the 13,602 genes that pass uniform quality control and recomputing the Benjamini–Hochberg adjustment over that universe rediscovers all 206 frozen candidates and adds 12 further genes, so the frozen list is a conservative subset of the QC-first analysis. Positive interaction effects (stronger infection response in Yurong1)

and negative effects (stronger response in Guiwei) were both well represented. At the pathway level, six Plant Reactome pathways transported through one-to-one *Oryza* orthology met the frozen discovery threshold: activation of the pre-replication complex, assembly of the pre-replication complex, DNA replication initiation, maturation, and circadian rhythm, all with positive enrichment, and jasmonic acid signaling with negative enrichment. Together with the genes, these six pathways, the apegglm-shrunken signed signature weights over the 206 genes, and the conditional transcript-usage results (Section 2.9) were frozen before any external data were opened.

2.5 None of the 18 exploratory candidates met the genome-wide interaction criterion

Because the exploratory candidates were selected before any interaction P -values existed, the registered protocol included a formal audit of all 18 against the genome-wide analysis (Figure 2B; Table 2). None met the genome-wide discovery threshold. Two questions must be kept distinct here. The audit tests the cultivar-dependent response claim under the interaction model; it does not retest the within-cultivar infection responses for which most of the 18 were originally selected, which is a different estimand requiring per-cultivar contrasts under the confirmatory pipeline and is planned as follow-up work. For the subset explicitly nominated as cultivar-skewed in the exploratory stage, the interaction test is the appropriate audit; for the remainder, the result below should be read as failure of the newly applied interaction criterion, not as disproof of their original infection-response claims. The strongest legacy candidate, LITCHI001510, reached $q = 0.080$; the EF-hand calcium-binding gene LITCHI019519, whose selected-gene contrast of -3.4 had appeared most dramatic in the exploratory stage, reached only $q = 0.106$ despite a re-estimated interaction effect of -3.33 ; and the median genome-wide q across the 18 candidates was 0.47 . Two candidates, the osmotin/thaumatin-like gene LITCHI009301 and LITCHI028401, additionally failed uniform exon-mappability control, meaning their expression estimates cannot be attributed unambiguously to a single locus. The audit does not show that these genes are biologically inert; several retain large, direction-consistent point estimates. It shows that the evidence available in this cohort cannot distinguish them from the noise floor of a genome-wide search, which is precisely the distinction the exploratory stage could not make.

2.6 Sixteen genes and one pathway passed all internal robustness gates

Each of the 206 frozen genes was required to retain its interaction direction and significance under four pre-declared perturbations of the analysis: independent Salmon-based quantification (196 of 206 passed), edgeR quasi-likelihood testing in place of the primary framework (19 passed), a counts-per-million expression filter (205 passed), and all twelve leave-one-library-out refits (125 passed). Eighteen genes passed all four gates jointly, and 16 remained after an additional observed mapping-sensitivity gate that excluded candidates whose effect depended on ambiguously mapped reads (Figure 3A). The alternative-framework gate was by far the most stringent, and the frozen genome-wide edgeR results explain why. The two frameworks agree almost perfectly on estimation: interaction effects correlate at Pearson $r = 0.995$ across all 19,445 genes and $r = 0.999$ across the 206 candidates, sign agreement among the candidates is 100%, and every candidate has nominal edgeR $P < 0.05$. The disagreement is confined to significance calibration. The registered gate required genome-wide edgeR $q < 0.10$ with direction agreement, and edgeR's quasi-likelihood test with robust dispersion estimation admits only 24 genes genome-wide at that threshold (10 at $q < 0.05$), against 262 for the primary Wald analysis at $q < 0.05$. With three libraries per cell, the two dispersion-estimation strategies imply very different genome-wide significance sets even while agreeing on every point estimate. We therefore present the results as a hierarchy: 262 statistical candidates, 206 retained after quality control, 19 significant under both frameworks, and 16 passing the complete registered procedure; genes sur-

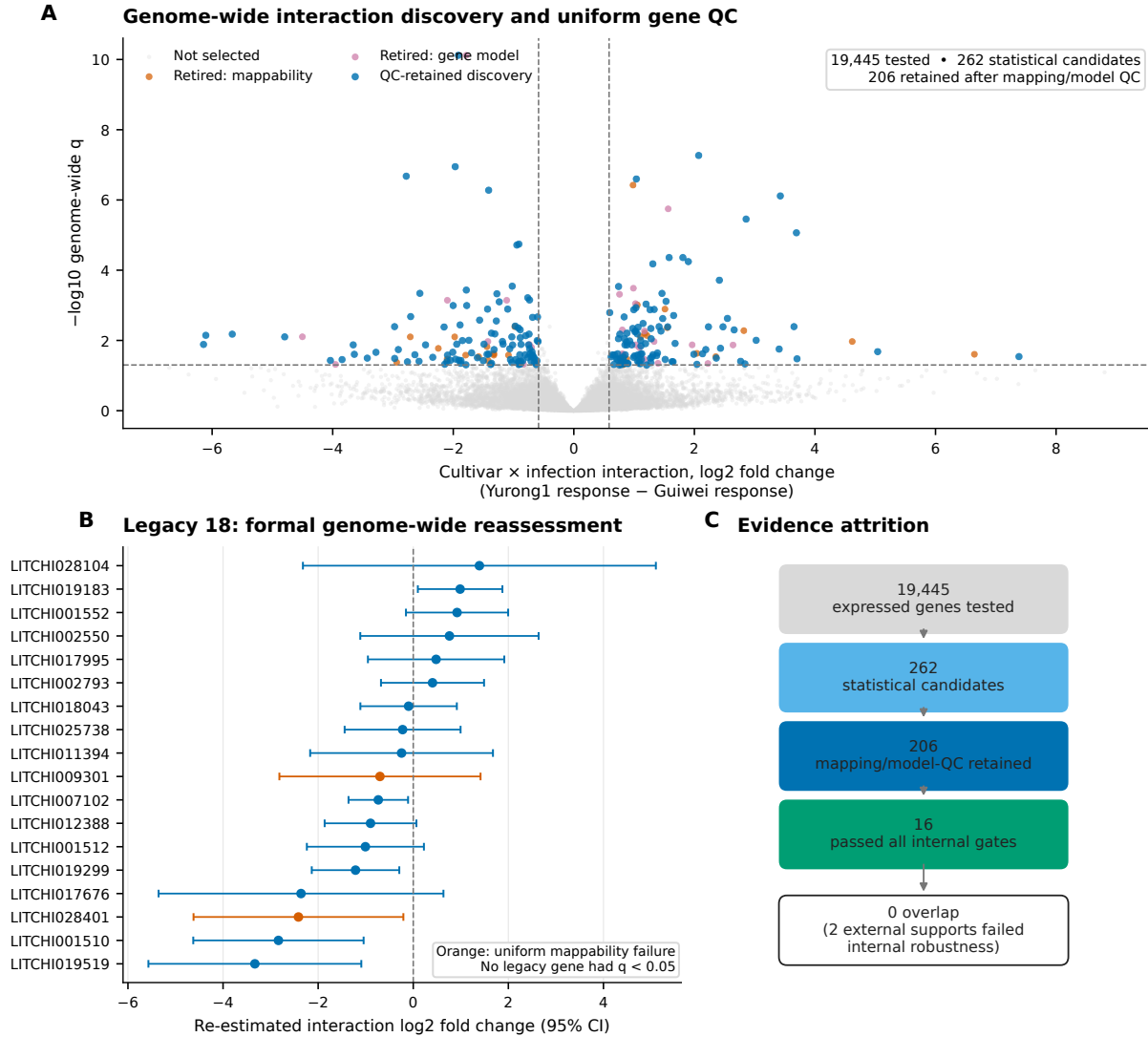


Figure 2: Genome-wide discovery and formal reassessment of the exploratory candidates. (A) Cultivar-by-infection interaction effects across 19,445 expressed genes; 262 genes met the statistical threshold and 206 were retained after uniform mapping and gene-model quality control. (B) Genome-wide re-estimation of the 18 exploratory (legacy) candidates with 95% confidence intervals; orange marks uniform-mappability failures. No legacy candidate met the genome-wide interaction criterion. (C) Evidence attrition from tested genes to internally robust candidates.

viving both frameworks constitute the discovery set’s stable core, and the wider 206-gene list should be read as one framework’s candidates rather than as settled discoveries. Of the six frozen pathways, only circadian rhythm passed its four conjunctive gates (competitive and rotation-based testing, leading-edge deletion, and an expression- and length-matched null); the other five all failed the rotation-based self-contained test, and the maturation set additionally failed leading-edge deletion, indicating enrichment that was not stable across testing frameworks.

2.7 The locked external evaluation supported two genes and contradicted five

The primary external estimand, fixed before unlock, was the resistant-minus-susceptible cultivar-by-infection interaction at 24 h in the PRJNA450886 pericarp time course, tested for each frozen candidate with Benjamini–Hochberg adjustment across the full candidate-by-contrast family, an effect threshold of $\log_2(1.5)$, confidence intervals excluding zero, and required agreement with the discovery direction. Of the 206 frozen genes, 184 were measurable in the external cohort. Two met every criterion and are cross-context supported: LITCHI005518, annotated as an alpha-xylosidase involved in xyloglucan remodeling of the cell wall, and LITCHI001963, an uncharacterized gene with a one-to-one *Oryza* ortholog. Five genes, including a beta-glucosidase and a Fe(II)/2-oxoglutarate-dependent dioxygenase, passed the magnitude and significance criteria with the *opposite* direction and are recorded as contradictory; 177 were unsupported at the prespecified thresholds and 22 not testable (Figure 3C; Table 3). At the level of continuous estimates, discovery and external interaction effects across the 184 measurable candidates were uncorrelated (Pearson $r = -0.05$, 95% CI -0.19 to 0.10 ; Spearman $\rho = -0.04$), with sign concordance of 51.6% (95 of 184), indistinguishable from chance.

The two supported genes are not among the 16 internally robust genes, and none of the 16 internally robust genes achieved external support (Figure 3B). This non-overlap is, by itself, nearly uninformative: for sets of 16 and 2 genes drawn from 206 candidates, the expected overlap under independence is 0.16 genes and the probability of zero overlap is 0.85, so the disjoint sets neither demonstrate nor refute an association between internal stability and cross-context transport. The absence of continuous concordance reported above is the more informative observation. We emphasize that the external cohort differs in tissue (pericarp versus leaf), resistant comparator (Heiye versus Yurong1), and design (time course versus single time point), so absence of support is expected wherever a response is tissue- or genotype-specific, and support that is observed indicates transport across contexts rather than replication of the original experiment. Because Yurong1 and Heiye have not been placed on a common resistance scale relative to Guiwei, we interpret the primary external test as cross-cultivar response transport rather than validation of a resistance phenotype.

2.8 Pathway and signature transport was weak and strongly context-dependent

None of the six frozen pathways, including the internally robust circadian-rhythm pathway, passed the external pathway gates in PRJNA450886 (Figure 4A). The frozen signed signature over the 206 discovery genes behaved in a strongly context-dependent way (Figure 4B). At the primary 24-h contrast the signature score was indistinguishable from zero (estimate 0.109, 95% CI -0.079 to 0.297 , $q = 0.349$), and at 48 h it was significantly *negative* (-0.240 , 95% CI -0.428 to -0.052 , $q = 0.043$), a sign reversal relative to discovery. In the generic-transfer study the signature tracked infection positively in leaf (0.107 , $q = 0.0015$) but negatively in fruit (-0.181 , $q < 0.001$), with a significant leaf-minus-fruit interaction (0.289 , $q < 0.001$). The signature therefore carries reproducible infection-associated information, but its sign and magnitude depend on tissue and time in ways that preclude its use as a context-free resistance marker. The exploratory cohort’s estimate is reported for completeness and is not confirmatory.

Table 2: Formal genome-wide audit of the 18 exploratory-stage candidates. Interaction effects are re-estimated genome-wide values (Yurong1 response minus Guiwei response, log2 scale). No candidate met the registered discovery threshold (genome-wide $q < 0.05$ and $|\log_2\text{FC}| \geq \log_2(1.5)$).

Gene	Exploratory annotation	Interaction log2FC (SE)	Genome-wide q	Mappability
LITCHI001510	Poorly characterized responsive gene	−2.83 (0.91)	0.080	Pass
LITCHI019519	EF-hand calcium-binding protein	−3.33 (1.14)	0.106	Pass
LITCHI019299	S-adenosylmethionine-related protein	−1.21 (0.47)	0.175	Pass
LITCHI007102	Phytosulfokine precursor	−0.74 (0.32)	0.238	Pass
LITCHI019183	Plant lipid-transfer protein	+0.99 (0.45)	0.273	Pass
LITCHI028401	Selected interaction candidate	−2.41 (1.13)	0.281	Fail
LITCHI012388	Heat-shock protein 70	−0.90 (0.49)	0.381	Pass
LITCHI001552	Cyclase-like stress-related protein	+0.92 (0.55)	0.430	Pass
LITCHI001512	Poorly characterized responsive gene	−1.01 (0.63)	0.457	Pass
LITCHI017676	S-adenosylmethionine-related protein	−2.36 (1.53)	0.478	Pass
LITCHI002550	Cytochrome P450	+0.76 (0.96)	0.762	Pass
LITCHI028104	Jacalin-like lectin	+1.39 (1.89)	0.785	Pass
LITCHI002793	Osmotin/thaumatin-like protein	+0.41 (0.55)	0.786	Pass
LITCHI017995	Metalloendoproteinase	+0.48 (0.73)	0.812	Pass
LITCHI009301	Osmotin/thaumatin-like protein	−0.70 (1.08)	0.816	Fail
LITCHI025738	Cyclase-like stress-related protein	−0.22 (0.62)	0.908	Pass
LITCHI011394	Pathogenesis-related protein	−0.25 (0.98)	0.940	Pass
LITCHI018043	Plant lipid-transfer protein	−0.10 (0.52)	0.954	Pass

Table 3: Cross-context supported and contradictory genes in the primary external evaluation (PR-JNA450886, pericarp, 24 h). Discovery effects are Yurong1-minus-Guiwei interaction log2 fold changes in leaf; external effects are Heiye-minus-Guiwei interaction log2 fold changes in pericarp with 95% confidence intervals. All q -values are Benjamini–Hochberg adjusted (discovery: genome-wide; external: across the frozen candidate-by-contrast family).

Gene	Annotation	Disc. log2FC	Disc. q	Ext. log2FC (95% CI)	Ext. q	Outcome
LITCHI005518	Alpha-xylosidase 1	+0.64	0.025	+0.67 (+0.32 to +1.02)	0.007	Supported
LITCHI001963	Uncharacterized; 1:1 <i>Oryza</i> ortholog	−1.41	0.024	−0.94 (−1.43 to −0.46)	0.007	Supported
LITCHI030761	Caffeoylshikimate esterase	−0.92	0.040	+1.31 (+0.78 to +1.85)	0.0002	Contradictory
LITCHI029534	Thiamine thiazole synthase 2	+1.48	0.002	−1.04 (−1.61 to −0.48)	0.010	Contradictory
LITCHI021860	Phosphatidylinositol 4-kinase gamma 2	−0.65	0.032	+0.97 (+0.39 to +1.54)	0.029	Contradictory
LITCHI008313	Beta-glucosidase 44	+0.90	0.045	−1.12 (−1.81 to −0.44)	0.034	Contradictory
LITCHI001700	Fe(II)/2-OG-dependent dioxygenase 4	+3.70	0.033	−2.00 (−3.24 to −0.77)	0.035	Contradictory

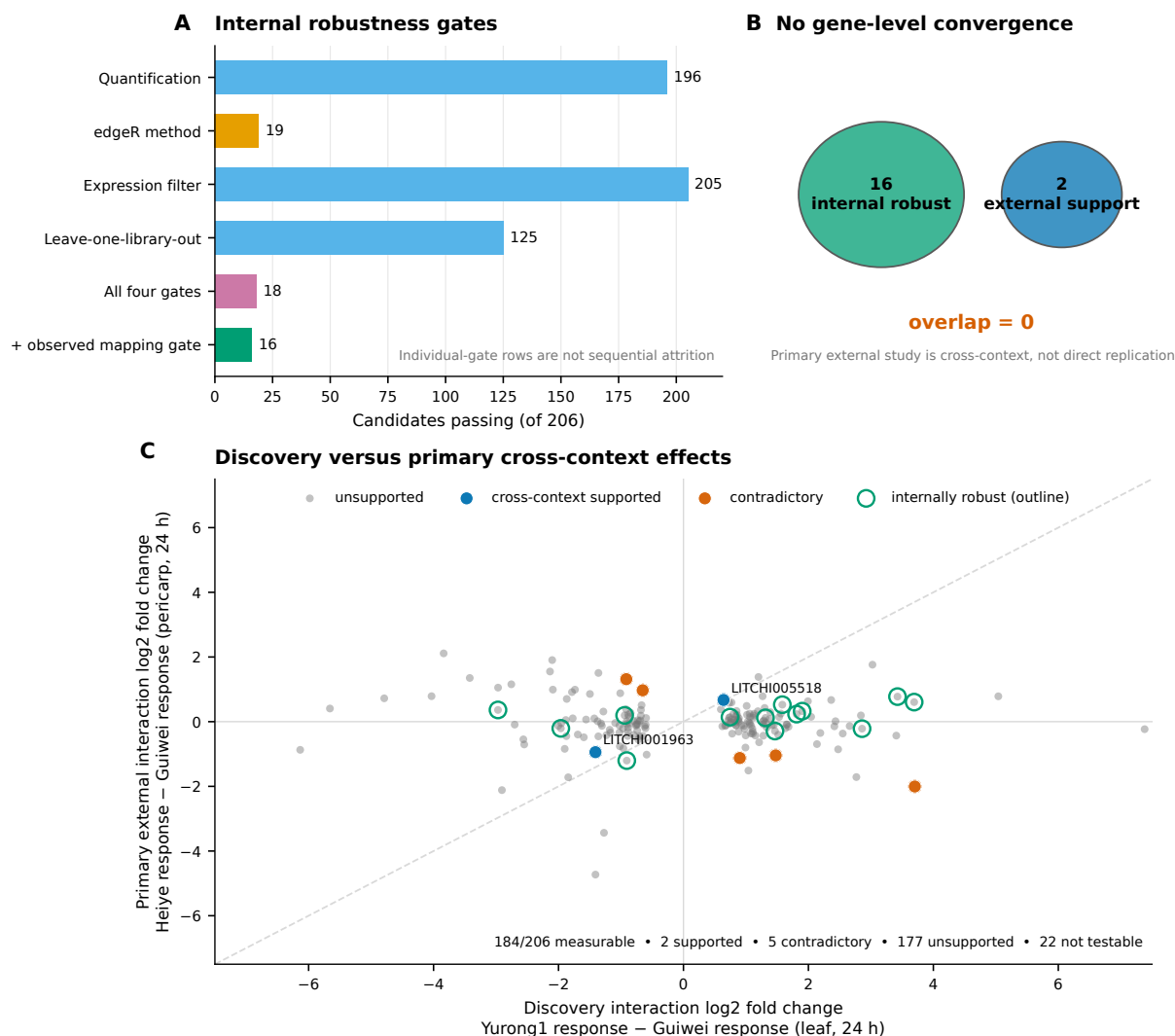


Figure 3: Internal robustness and the primary external evaluation. (A) Candidates passing each predeclared robustness gate, the conjunction of the four analytic gates, and the additional observed-mapping gate. (B) The internally robust set (16 genes) and the externally supported set (2 genes) do not overlap. (C) Discovery interaction effects against the primary cross-context interaction effects (Heiye response minus Guiwei response, pericarp, 24 h); supported, contradictory, and internally robust genes are highlighted. The diagonal is a visual reference only; equal effects across contexts are not expected.

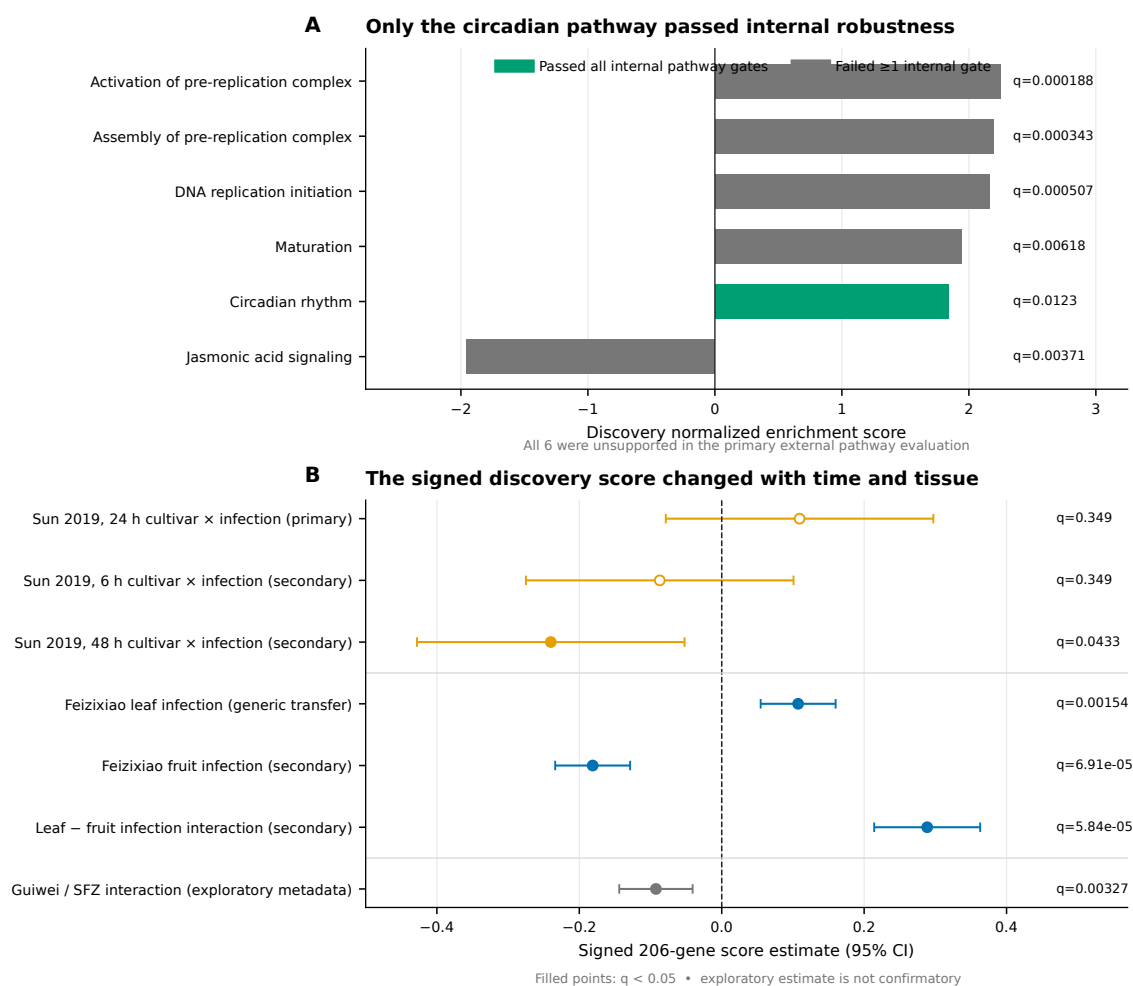


Figure 4: Frozen pathway and signature evaluation. (A) Discovery normalized enrichment scores for the six frozen pathways; only circadian rhythm passed all internal gates, and none was supported in the primary external evaluation. (B) The frozen signed 206-gene signature across the seven locked external and exploratory contrasts; filled points denote $q < 0.05$.

2.9 Conditional transcript-usage discovery remained internal

Differential transcript usage was analyzed under a conditional gate registered in advance: stage-wise testing had to yield interpretable gene- and transcript-level error control, and no post hoc substitute was permitted if the gate failed. The gate passed, and 225 transcript-usage events across 152 genes were frozen from 15,790 tested transcripts (Figure 5A). External follow-up preserved the role separation of the cohorts (Figure 5C): in the primary cross-context study, 125 events were measurable but unsupported at the prespecified thresholds and 100 were not testable; in the generic-transfer study, annotation incompatibilities left all 225 untestable; and in the exploratory cohort, 4 events were supported and 5 contradictory; we report these outcomes descriptively and do not count them as primary external support, because that cohort’s metadata could not be standardized. Transcript-level rewiring between cultivars therefore remains an internally supported, externally unconfirmed layer of the resource.

2.10 Orthogonal evidence: informative nulls and reference-ecosystem gaps

The orthogonal evidence classes returned uniformly negative or non-testable results, and we report them as such rather than converting them into support. Computational annotation assigned family-level functional labels to 150 of the 206 candidates and left 56 unannotated, but no candidate met the registered high-confidence rule requiring two independent evidence classes with at least 70% supported coverage and no architecture conflict, largely because precomputed InterPro domain architectures were unavailable for all 206 proteins in this non-model species (Figure 5B). Promoter-motif discovery, which in the exploratory stage had suggested enrichment of stress-associated elements against randomized backgrounds, was repeated under far stricter conditions: 927 JASPAR plant position-weight matrices tested in strand-aware 1-kb and 2-kb promoter windows against 100 expression- and GC-matched genomic background sets, with AME as the primary test, FIMO as sensitivity, and cognate transcription-factor expression required. No motif passed in more than 4 of 100 matched backgrounds against a required 80, and none passed FIMO sensitivity (Figure 5D). The contrast with the exploratory result is instructive: matched genomic backgrounds absorb the compositional signal that simple randomized backgrounds mistake for enrichment. Small-RNA coherence could not be tested at all, because the frozen PmiREN reference contained no exact *Litchi* entries; substituting study-derived novel miRNAs after seeing the gate outcome would have violated the protocol, so the class is recorded as not testable. Finally, published literature reusing any accession analyzed here, including the primary external study itself [5], was registered as prior interpretation rather than independent support.

2.11 Deterministic integration: an empty top tier, twelve Tier B genes, and visible retirements

The registered tier rules integrate all evidence layers without discretion. Tier A required discovery, complete internal robustness, primary cross-context support, clean mapping and annotation, and at least one attributable orthogonal class; Tier B required discovery and full internal robustness with partial or non-testable external evidence; Tier C was reserved for a robust, externally supported pathway. Applied to the 268 frozen entities (262 statistical gene candidates and 6 pathways), the rules assigned no entity to Tier A or Tier C, 12 genes to Tier B, 191 entities to exploratory status, and 65 to retirement (Figure 5E). The retirements comprise the 56 statistical candidates removed by uniform mapping or gene-model control, the 5 genes contradicted for cross-context directional transport, and 4 candidates with candidate-level observed-mapping or annotation failures. These categories differ in kind, and the released tables preserve the distinction: mapping and annotation failures undermine the discovery itself, whereas an opposite direction in a different tissue and cultivar contrast may reflect genuine context specificity rather than an invalid discovery in the original context.

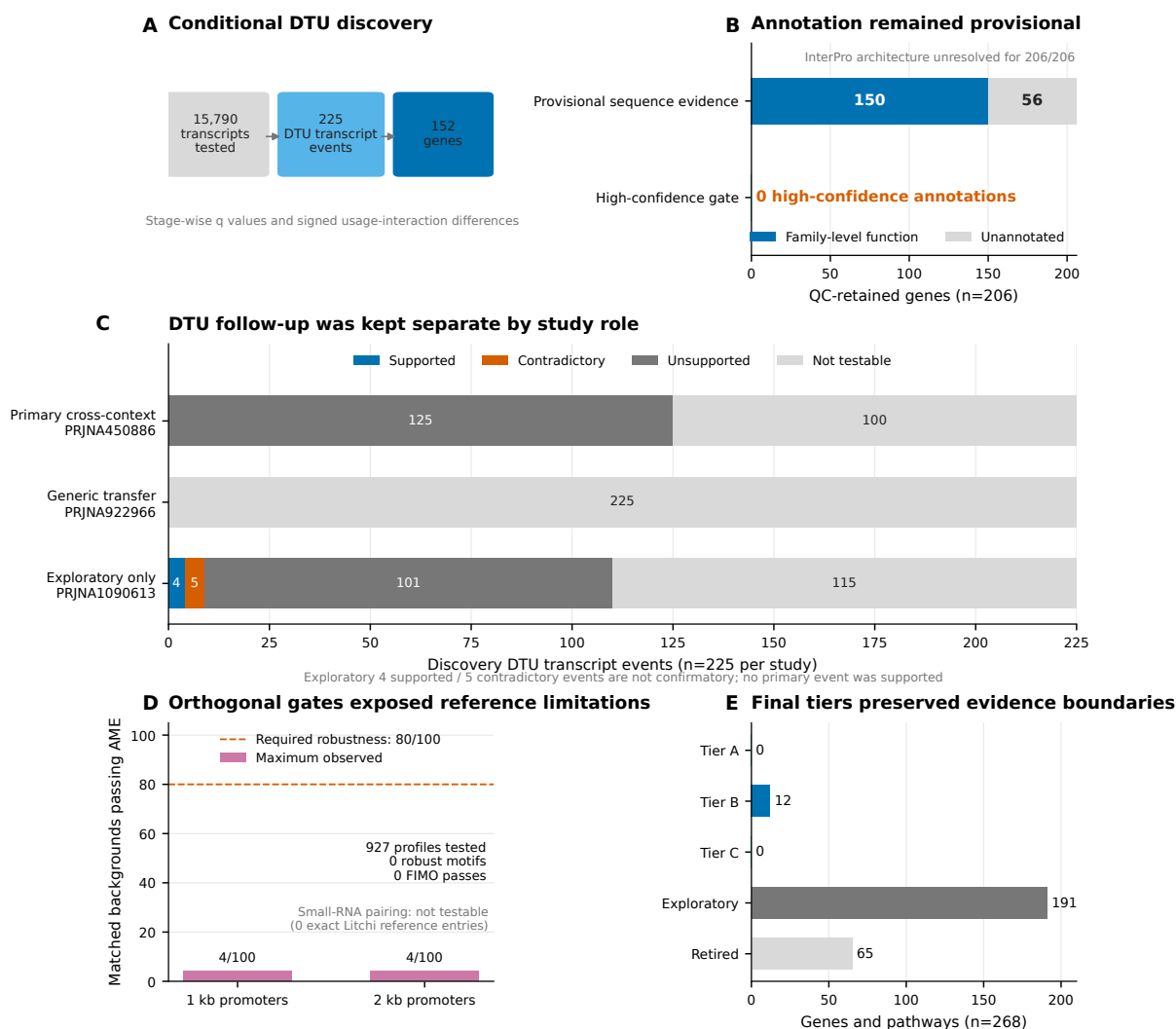


Figure 5: Transcript usage, orthogonal evidence, and final tiers. (A) Conditional differential-transcript-usage discovery. (B) Annotation status of the 206 QC-retained genes; no gene met the high-confidence two-class rule. (C) External DTU follow-up separated by locked study role. (D) Promoter-motif robustness against 100 expression- and GC-matched backgrounds and the small-RNA reference gate. (E) Deterministic final tiers over all 268 frozen entities.

Table 4: The twelve Tier B genes: internally robust genome-wide statistical candidates with direction-consistent but sub-threshold, or non-measurable, primary external evidence. Discovery effects are interaction log2 fold changes (Yurong1 response minus Guiwei response); positive values indicate a stronger infection response in Yurong1. External columns give the primary 24-h pericarp estimate with 95% confidence interval and family-adjusted q ; NM denotes genes not measurable in the external cohort.

Gene	Family-level annotation	Disc. log2FC	Genome-wide q	Ext. log2FC (95% CI)	Ext. q
LITCHI014613	Sugar transporter ERD6-like 16	+3.69	8.7×10^{-6}	+0.60 (−0.04 to +1.25)	0.27
LITCHI015982	WAT1-related protein	+3.43	7.8×10^{-7}	+0.77 (+0.07 to +1.48)	0.18
LITCHI005805	Phosphomannomutase/phosphoglucomutase	+2.07	5.5×10^{-8}	NM	—
LITCHI028926	Exonuclease 1	+1.90	5.7×10^{-5}	+0.32 (−0.63 to +1.28)	0.83
LITCHI016077	Serine carboxypeptidase 24	+1.81	4.4×10^{-5}	+0.23 (−1.05 to +1.51)	0.89
LITCHI013915	1,4-alpha-glucan-branching enzyme 3	+1.58	4.4×10^{-5}	+0.53 (+0.10 to +0.95)	0.13
LITCHI021546	Homeobox-DDT domain protein RLT3	+1.31	6.6×10^{-5}	+0.12 (−0.30 to +0.54)	0.83
LITCHI022605	Ribonuclease J	+0.74	2.9×10^{-4}	+0.14 (−0.34 to +0.62)	0.83
LITCHI028717	BAG-family chaperone regulator 4	−0.91	1.8×10^{-5}	−1.20 (−2.52 to +0.12)	0.29
LITCHI008721	Unannotated protein	−1.91	7.9×10^{-11}	NM	—
LITCHI015740	Putative lipid-transfer protein DIR1	−1.96	1.2×10^{-7}	−0.21 (−1.10 to +0.69)	0.88
LITCHI010877	Cytochrome P450 81Q32	−2.77	2.1×10^{-7}	NM	—

Four internally robust genes, including a probable RPP8-like disease-resistance protein, were measurable but unsupported at the prespecified thresholds in the primary external evaluation and were therefore held at exploratory status rather than promoted. The two cross-context-supported genes likewise remain exploratory because they failed internal robustness. All contradictory and failed outcomes are released alongside the positive ones. We also note that Tier A was difficult to reach by construction in the current reference ecosystem: with the small-RNA class structurally untestable, no motif able to pass the family-wise rule, accession-linked literature excluded as independent support, and the two-class annotation rule unattainable without precomputed domain architectures, the orthogonal-class requirement could not realistically be satisfied by any gene. The empty top tier therefore reflects this structural constraint as much as the expression evidence itself.

The 12 Tier B genes are listed in Table 4. Nine were measurable in the primary external cohort, and all nine were direction-consistent with discovery, although none met the full support criteria. Their annotations cluster around carbohydrate and cell-wall metabolism (a phosphomannomutase/phosphoglucomutase, a chloroplastic 1,4-alpha-glucan-branching enzyme, an ERD6-like sugar transporter, a serine carboxypeptidase), membrane transport (a WAT1-related protein), lipid signaling (a DIR1-like lipid-transfer protein, the *Arabidopsis* ortholog of which is required for systemic acquired resistance), programmed-cell-death regulation (a BAG-family chaperone regulator), specialized metabolism (cytochrome P450 81Q32), RNA metabolism (ribonuclease J, exonuclease 1), chromatin-associated regulation (a homeobox-DDT protein), and one entirely unannotated gene.

3 Discussion

The central finding of this unified analysis is methodological and biological at once: in the lychee–*P. litchii* pathosystem, the genes that look most impressive in a single-cohort exploratory analysis and the genes that survive formal genome-wide inference are almost entirely different sets. The 18 exploratory candidates were chosen for large fold changes and defense-plausible annotations, yet none reached genome-wide significance for the cultivar-by-infection interaction, and two could not even be attributed unambiguously to a single locus. Conversely, the 16 genes that passed every internal robustness gate, and the 12 that reached Tier B, are dominated not by canonical pathogenesis-related families but by carbohydrate and cell-wall metabolism,

transport, lipid signaling, and RNA metabolism, including one gene with no annotation at all. Neither observation would have been visible without running both stages, and together they quantify a selection effect that is likely to operate in any small-replicate transcriptomic study of this kind. One qualification applies throughout: the audit changed not only the statistical machinery but, for many of the 18 genes, the scientific question, from within-cultivar responsiveness to cultivar-dependent response, so the comparison measures survival under a stricter and partly different criterion rather than direct refutation of the original claims.

Why did the exploratory candidates fail the interaction audit? Three mechanisms, all anticipated by the reviewers of the exploratory report, are sufficient to explain the outcome. First, selection on observed fold change in a three-replicate contrast preferentially captures genes with high sampling variance, and their apparent effects regress when the full uncertainty is modeled: LITCHI019519's selected-gene contrast of -3.4 looked decisive, but its genome-wide interaction test carried a standard error of 1.14 and a q -value of 0.106. Second, per-cultivar differential expression followed by informal comparison is not an interaction test; genes can differ dramatically in within-cultivar significance while their responses do not differ significantly between cultivars. Third, two candidates failed uniform-mappability control, a failure mode invisible to any analysis that trusts a single quantification path. None of this implies the exploratory candidates are biologically irrelevant; several retain large direction-consistent estimates and remain reasonable targets for quantitative PCR. Their evidentiary status, however, is now stated correctly: unconfirmed hypotheses, not validated markers.

The annotations of the genes that survive scrutiny suggest a recognizable, although still hypothetical, theme. Annotations related to cell-wall and sugar metabolism appear at every supported evidence level: among the Tier B genes, a phosphomannomutase/phosphoglucomutase, a starch-branching enzyme, an ERD6-like sugar transporter, and a serine carboxypeptidase; and among the two cross-context-supported genes, an alpha-xylosidase acting on xyloglucan. We emphasize that this is a qualitative clustering of family-level annotations over small gene sets, assembled from different genes at different evidence levels, and not a formal over-representation result; no enrichment test was performed, and several annotations are provisional. The pattern is nonetheless consistent with the emphasis on sugar metabolism and cell-wall reinforcement in prior comparisons of resistant and susceptible lychee cultivars [5], and with the general role of apoplastic sugar status and wall remodeling in oomycete resistance. The Tier B set also contains a DIR1-like lipid-transfer protein, whose *Arabidopsis* counterpart is a mobile signal in systemic acquired resistance, and a BAG-family chaperone regulator; BAG proteins act as HSP70 co-chaperones within the chaperone networks that shape plant immune responses and immune-associated cell death [10]. The strong negative discovery enrichment of jasmonic acid signaling, although it did not survive external evaluation, is likewise a plausible axis of cultivar difference given the jasmonate-associated promoter elements suggested in the exploratory stage and the known antagonism between jasmonate and salicylate defense programs. We advance all of these as prioritized, testable hypotheses; the tier system exists precisely to prevent us from advancing them as more.

The external evaluation delivers a second structural lesson. The internally robust and externally supported sets did not overlap, but this disjunction carries almost no information on its own: with 16 and 2 genes selected from 206 candidates, the expected overlap under independence is 0.16 and zero overlap occurs with probability 0.85. The informative observations are continuous ones: discovery and external interaction effects were uncorrelated across the 184 measurable candidates, the frozen signature reversed sign between 24 h and 48 h in the external time course, and the same signature tracked infection in opposite directions in leaf and fruit of the transfer cohort. The straightforward reading is that the cultivar-dependent component of the early infection response is substantially tissue- and time-specific, so that a candidate's failure to transport from leaf at 24 h to pericarp at 48 h is expected behavior rather than evidence against the discovery. This has a practical corollary for breeding applications: expression-based resistance markers derived from one tissue and time point should not be assumed portable, and marker development for this pathosystem will require

validation in the tissue and developmental window where deployment is intended.

The orthogonal layers contribute mainly disciplined null results, and we consider their inclusion one of the more useful aspects of the resource. The promoter-motif contrast between stages is consistent with background choice having produced the exploratory enrichment signal, but it does not isolate that cause: the two analyses differed simultaneously in foreground genes, motif definitions, scoring procedure, and robustness criteria, so the defensible statement is that the exploratory motif findings did not survive a more stringent and methodologically different analysis. A controlled comparison holding everything but the background construction fixed remains to be done. The annotation and small-RNA outcomes expose reference-ecosystem gaps (no precomputed InterPro architectures and no *Litchi* entries in the frozen miRNA reference) that bound what any computational study of this species can currently claim, and that will silently inflate confidence in studies that do not check them. Reporting these gaps as non-testable classes, rather than quietly substituting weaker evidence, is what keeps the final tiers interpretable.

Several limitations qualify all of the above. The deposited designs are small, with three libraries per cell, and the independence of those libraries at the level of source trees, pooling, and extraction is unreported, so our q -values are conditional on deposited-library independence. The public metadata also leave the inoculation protocol underspecified, including the inoculum form (mycelia, sporangia, or zoospores), the developmental stage of the sampled leaves, and whether the tissue was wounded before inoculation, so all conclusions are additionally conditional on the depositors' experimental design. All quantification depends on a single reference assembly, and although uniform and observed mapping checks removed the clearest failure modes, residual reference bias cannot be excluded. The primary external cohort is deliberately heterogeneous with respect to the discovery cohort, which makes support conservative and non-support ambiguous; no genuinely independent replication cohort (same tissue, same time, different biological material) exists in the public record for this design. Unsupported external outcomes additionally conflate limited power, context dependence, and true absence of effect: with three libraries per design cell, minimum detectable interaction effects are large, and a simulation-based power analysis quantifying them has not yet been performed. And the entire study is computational: no candidate, including the Tier B set, has been perturbed experimentally. The natural next steps are quantitative PCR of the Tier B genes in independently infected material of both discovery cultivars, functional tests of the alpha-xylosidase and DIR1-like candidates, and, for any regulatory hypothesis, direct assays such as promoter-reporter or chromatin-binding experiments rather than further motif counting.

4 Conclusions

By carrying a single pathosystem from exploratory candidate discovery through a prospectively registered genome-wide discovery framework with locked cross-context external evaluation, this study establishes which claims about cultivar-dependent responses of lychee to *P. litchii* the current public data can and cannot support. It can support a set of 206 QC-retained genome-wide statistical interaction candidates, of which 19 are significant under both statistical frameworks, a stable core of 16 internally robust genes, 12 Tier B candidates whose annotations include cell-wall, sugar-metabolism, lipid-signaling, and RNA-metabolism functions, and 2 genes with cross-context support in an independent tissue and cultivar contrast. It cannot support the original 18 exploratory candidates as validated markers, any pathway- or signature-level claim of context-free transport, or any regulatory-element claim under matched-background testing. All frozen results, including nulls, contradictions, and retirements, are released with the analysis code and checksummed source data, so that the next stage of experimental validation can begin from an honest baseline.

5 Materials and Methods

5.1 Study design, registration, and evidence vocabulary

The confirmatory stage was governed by a time-stamped protocol, with amendments logged, distributed with the accompanying repository. The registration chronology is as follows. The exploratory stage was completed and publicly documented first [9], so the discovery dataset had already been analyzed, under different models, before registration. Protocol version 1.0.0 was frozen on 18 August 2026, with its SHA-256 hash (ef668486...) recorded alongside the protocol text, before any confirmatory expression outcome was computed. The amendment log records 20 outcome-blind amendments (clarifications, implementation corrections, and optimizations, all time-stamped before any discovery or external expression outcome was viewed) and 3 post-outcome amendments confined to technical quality control and manifest verification, each marked as such. The primary external study had been published previously [5], and its headline biological conclusions were known to the analyst before registration; dataset roles and all thresholds were nevertheless fixed before any external count matrix was processed. Because all cohorts are public, the staged external unlock is a workflow-enforced computational safeguard, not blinding in the experimental sense. The protocol fixed dataset roles (Table 1), statistical models, discovery and evaluation thresholds, conditional gates, empty-set behavior, and the vocabulary used throughout this paper: *discovery* refers only to the locked genome-wide interaction analysis in PRJNA830488; *robustness* refers only to predeclared sensitivity analyses within that cohort; *cross-context support* denotes a passing frozen external test in a cohort that differs in tissue, comparator, or design, and is never described as replication; and *orthogonal* evidence classes (annotation, motifs, small RNA, literature) are reported separately from expression support and cannot substitute for it. No external outcome could modify frozen discovery membership, signature weights, or thresholds.

5.2 Exploratory-stage analyses (Stage 1)

The exploratory stage, reported in full elsewhere [9], analyzed the 12 GSE201243 libraries with FastQC v0.11.9 quality control, Trimmomatic v0.39 trimming, TopHat v2.0.12/Bowtie2 v2.2.3 alignment to the SCAU_Lch_v2.0 assembly (GCA_019925255.1), HTSeq v2.0.2 union-mode counting, and DESeq2 v1.42.0 per-cultivar Wald contrasts (design $\sim$ treatment) with Benjamini–Hochberg adjustment at $P < 0.05$ [11–16]. Candidate prioritization combined significance, fold-change magnitude, and annotation plausibility; structural, conserved-motif, and promoter analyses used TBtools-II v2.096, MEME v5.5.5, and PlantCARE [17–19]; and the promoter-background comparison tested observed *cis*-element counts in the 18 candidate promoters against 100,000 randomized sets of eighteen 2-kb sequences with lychee-like GC content, scoring exact motifs and reverse complements, with one-sided empirical P -values adjusted by Benjamini–Hochberg [20]. The selected-gene interaction contrast was computed as $\log_2\text{FC}(\text{Yurong1, infected versus mock}) - \log_2\text{FC}(\text{Guiwei, infected versus mock})$ over the prioritized genes and was explicitly labeled a prioritization effect size rather than a genome-wide test.

5.3 Data provenance and eligibility

National Center for Biotechnology Information and Gene Expression Omnibus metadata for all five cohorts were reconciled into a biological-unit registry (Supplementary S1). Because source-tree, pooling, harvest, and extraction independence were not reported for the discovery libraries, all inferential q -values are conditional on deposited-library independence. The primary model required at least three included libraries in every cultivar-treatment cell.

5.4 Confirmatory preprocessing and quality gates

Host (SCAU_Lch_v2.0) and pathogen references were combined with prefixed contig names to absorb pathogen-derived reads. Paired reads were checked with FastQC, trimmed with fastp [21], aligned with STAR two-pass mapping [22], counted at the gene level with featureCounts [23], and independently quantified with Salmon [24]. Libraries were excluded if fewer than 10 million read pairs survived trimming or fewer than 40% of reads aligned uniquely; all 12 discovery libraries passed. GenMap-based exon-uniqueness scores [25] and candidate-level observed-mapping checks were fixed before discovery and used for the uniform mapping and observed mapping-sensitivity gates.

5.5 Genome-wide discovery model

DESeq2 [16] fitted gene counts with the design $\sim \text{cultivar} + \text{treatment} + \text{cultivar}:\text{treatment}$ (reference levels Guiwei, mock). The primary coefficient, `cultivarYurong1:treatmentinfected`, estimates the difference in infected-minus-mock response between Yurong1 and Guiwei. Genes required counts of at least 10 in at least three libraries; statistical candidates required genome-wide Benjamini–Hochberg $q < 0.05$ and $|\log_2\text{FC}| \geq \log_2(1.5)$ [20]. The registered rule tests the point null of zero interaction and applies the effect threshold to the observed estimate; a composite-null test of $|\beta| \leq \log_2(1.5)$ would be more stringent and was not part of the registered protocol, which is one reason we label the output statistical candidates. Unshrunk maximum-likelihood estimates are reported throughout; `apeglm`-shrunk interaction estimates [26] were used only to define the signed signature weights over the 206 frozen genes. Mappability and gene-model metrics were computed for all 19,445 tested genes; as a sensitivity check, recomputing the Benjamini–Hochberg adjustment over only the 13,602 QC-eligible genes rediscovered all 206 frozen candidates plus 12 additional genes, confirming that applying quality control after significance testing was conservative here.

5.6 Pathway and transcript-usage discovery

Plant Reactome pathway memberships [27] were transported to lychee through one-to-one reciprocal-best DIAMOND protein matches to *Oryza* [28], and pathway discovery applied the frozen genome-wide interaction statistic to this fixed mapping with camera, roast, and fgsea testing plus leading-edge-deletion and expression/length-matched null gates for robustness [29–31]. Differential transcript usage used Salmon transcript abundances analyzed with DRIMSeq and DEXSeq under stageR stage-wise error control [32–34]; the conditional inferential gate for interpretable two-level error control was registered in advance, and no substitute analysis was permitted had it failed.

5.7 Internal robustness gates

Each frozen gene was retested under Salmon-based gene-level quantification, edgeR quasi-likelihood interaction testing [35], a counts-per-million expression filter, and all leave-one-library-out refits, with predeclared criteria combined conjunctively; no additive scoring was used. The Salmon gate required direction agreement and an absolute effect difference of at most 0.5 log₂ units; the edgeR analysis used the identical design matrix and interaction coefficient with TMM normalization, robust dispersion estimation (`estimateDisp` and `glmQLFit` with robust settings), and `glmQLFTest`, with Benjamini–Hochberg adjustment computed genome-wide over all 19,445 genes and a gate of genome-wide $q < 0.10$ plus direction agreement; the expression-filter gate refit the primary model after a counts-per-million filter; and the leave-one-library-out gate required direction agreement in all 12 refits and $q < 0.10$ in at least 10 of 12. The observed mapping-sensitivity gate

removed candidates whose effect depended on ambiguously mapped reads. Frozen pathways required agreement across camera, roast, leading-edge deletion, and the matched-null comparison.

5.8 External frozen evaluation

PRJNA450886 was analyzed with the design $\sim \text{cultivar} \times \text{treatment} \times \text{time}$, with time encoded as a categorical factor and 24 h as the reference level, so that the cultivar-by-treatment coefficient estimates the 24-h interaction and the three-way terms give the 6-h and 48-h contrasts; the primary estimand was the resistant-minus-susceptible cultivar-by-infection interaction at 24 h, with 6-h and 48-h contrasts secondary. Frozen candidate tests used Benjamini–Hochberg adjustment across the full candidate-by-contrast family, an absolute effect threshold of $\log_2(1.5)$, confidence intervals excluding zero, and discovery-direction agreement; opposite-direction passes were recorded as contradictory. PRJNA922966 was analyzed with $\sim \text{tissue} + \text{treatment} + \text{tissue:treatment}$ for generic-transfer tests, and PRJNA1090613 outcomes were confined to exploratory status by protocol. The frozen signed signature was evaluated as a library-level score with the same family-wise adjustment.

5.9 Annotation, motif, small-RNA, and literature classes

Representative proteins were chosen by longest coding sequence with lexicographic tie-breaking. Reviewed Swiss-Prot (release 2026_02) DIAMOND matches [36], targeted InterPro/Pfam domain assignments [37], and one-to-one *Oryza* orthology formed three separate annotation classes; the high-confidence label required at least two classes, at least 70% supported coverage, and no architecture conflict. Promoter-motif analysis tested 927 JASPAR plant position-weight matrices [38] in strand-aware 1-kb and 2-kb promoter windows against 100 expression- and GC-matched background sets, with AME as the primary test, FIMO as sensitivity [39,40], robustness requiring passes in at least 80 of 100 backgrounds in both windows, and cognate transcription-factor expression required for attribution. The PmiREN bulk archive [41] was audited before small-RNA analysis; because it contained no exact *Litchi* entries, the class was frozen as not testable and no study-derived reference was substituted. Published studies reusing any analyzed accession were registered as prior interpretation.

5.10 Tier rules and reproducibility

Tier A required discovery, complete internal robustness, primary cross-context support, clean mapping and annotation, and at least one attributable orthogonal class; Tier B required discovery and complete internal robustness with partial or non-testable primary external evidence; Tier C was reserved for a robust, externally supported pathway. Contradictory external direction or mapping/annotation failure retired an entity, and an empty Tier A was an allowed outcome. All stages ran under Snakemake workflows [42] enforcing staged discovery, a controlled external unlock, verified cleanup, and SHA-256 manifests over inputs and outputs; the resolved software environment, synthetic test fixtures, per-figure source-data tables, and the amendment log are distributed with the repository. All five main figures are regenerated from frozen result tables by a single script that recomputes and asserts every headline count.

Declarations

Funding: This work received no external funding. **Conflicts of interest:** The author declares no conflicts of interest. **Author contributions:** E.Z. conceived the study, performed all analyses, and wrote the manuscript.

Data and code availability

All analyzed data are public: PRJNA830488 (GSE201243, discovery), PRJNA450886 (primary external), PRJNA922966 (GSE222651, generic transfer), PRJNA922965 (GSE222650, small RNA), and PRJNA1090613 (GSE262200, exploratory). The registered protocol, amendment log, complete result tables (all discovery statistics, robustness manifests, external tests, pathway/signature tests, transcript-usage results, annotation and orthology evidence, and the contradictory-results file), and the workflow code accompany this repository; a permanently archived release with a DOI, matching the submitted manuscript, will accompany the revised submission. Exploratory-stage scripts and supplementary tables are available with the original report [9].

Supplementary information

Supplementary tables S1–S13 provide the biological-unit registry, per-library quality control, all 19,445 discovery statistics, robustness manifests, all external frozen tests, pathway and signature tests, transcript-usage results, annotation and orthology evidence, the small-RNA gate record, all 752,724 motif-background tests, the accession-aware evidence registry, the script and environment inventory, and the protocol amendment log. Every main figure has a tab-separated source-data file.

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